Worked Answer Key

Page 1. 1. $500 + 200 = 700$. 2. $8,000 - 3,000 = 5,000$.

Page 2. 1. $37 + 14 = 51$. 2. $10 \times 4 = 40$.

Page 3. 1. $400 \times 20 = 8,000$. 2. $50 \times 20 = 1,000$ seats.

Page 4. 1. $200 + 300 + 400 = 900$. 2. $4,800 \div 60 = 80$.

Page 5. 1. $(400 - 1) + 268 = 667$. 2. $2,500 + 1,800 + (-3) = 4,297$.

Page 6. 1. $700 - 300 + 2 = 402$. 2. $2,796 + 204 = 3,000$, then $+2,000 = 5,000$; answer $2,204$.

Page 7. 1. $25 \times 16 = 100 \times 4 = 400$ (or $25 \times (8 + 8) = 200 + 200$). 2. $48 \times 5 = (48 \times 10) \div 2 = 240$.

Page 8. 1. $36 \times 100 = 3,600$, so $3,600 \div 9 = 400$. 2. Cancel a zero: $240 \div 6 = 40$.

Page 9. 1. $6,000 + 2,000 = 8,000$; low because the omitted hundreds and tens are positive. 2. $9,000 - 4,000 = 5,000$; low because the subtracted amount was rounded down and the minuend down.

Page 10. 1. $600 \times 7 = 4,200$, so $4,284$ is reasonable. 2. $5,000 \div 5 = 1,000$; 99 is not reasonable. (Exact answer is 990.)

Page 11. 1. Multiplication: estimate $6 \times 20 = 120$; exact $6 \times 24 = 144$. 2. Subtraction: estimate $240 - 100 = 140$; exact $238 - 97 = 141$.

Page 12. 1. $30 \times 20 + 10 \times 10 = 600 + 100 = 700$ approximately; exact $27 \times 19 = 513$, $14 \times 12 = 168$, total 681. 2. $8 \times 50 - 80 = 320$ approximately; exact $8 \times 46 = 368$, $368 - 75 = 293$.

Page 13. 1. Estimate $200 + 200 = 400$; exact $186 + 217 = 403$. 2. Estimate $2,300 - 900 = 1,400$; exact $2,304 - 879 = 1,425$.

Page 14. 1. Lower: $32 \times 24 = 768$; upper: $32 \times 29 = 928$ students. 2. 550 through 649.

Page 15. 1. Total $4 \times 38 = 152$; $152 \div 8 = 19$ cans per box, remainder 0. Plan: multiply, then divide. 2. Costs $3 \times 6 + 28 = 46$; $50-46=$4 remains.

Page 16. 1. $20 + 20 + 30 + 30 = 100$ estimate; exact $18 + 24 + 31 + 27 = 100$. 2. $5,000 + 5,000 + 5,000 = 15,000$ (nearest thousand); exact $14,770$.

Page 17. 1. About $3×15=$45; exact $14.95×3=$44.85. 2. About $40-8-13=$19; exact $40.00-$7.85-$12.60=$19.55.

Page 18. 1. $999 + 1,001 = (1,000 - 1) + (1,000 + 1) = 2,000$; also $999 + 1,000 + 1 = 2,000$. Both are equally efficient. 2. Round both: $50 \times 20 = 1,000$. Round only 19: $51 \times 20 = 1,020$; both are close, and $51 \times 19 = 969$ exactly.

Page 19. 1. Estimate $9 \times 400 - 700 = 2,900$; exact $9 \times 418 = 3,762$, $3,762 - 735 = 3,027$ occupied. 2. Estimate $300 + 400 + 300 + 200 + 400 = 1,600$, so yes, over $1,500$; exact total is $1,600$.

Page 20. 1. Estimate $3,200 + 1,900 - 900 = 4,200$; exact $3,248 + 1,876 = 5,124$, $5,124 - 925 = 4,199$. 2. Total $24 \times 36 = 864$; $864 \div 9 = 96$ markers per classroom. Estimate $20 \times 40 \div 10 = 80$, so 96 is reasonable.